

Differential Adventure: Spherical and Hyperbolic Geodesic Triangles

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1 Introduction: Geodesics

When I flew between Beijing and Los Angeles every summer as a kid, I was always baffled by the route that was taken by the plane. Why in the world would you go all the way north to Alaska and fly over Russia to go back down south to Beijing? From elementary and middle school, we learned that in Euclidean space, the straight line is the shortest path connecting two points. When we generalize the concept of straight lines to spheres or other curved surfaces, we step into the concept of “geodesics”. It turns out that the curves that are analogous to “straight lines” in Euclidean spaces are the great circles on a sphere, and the route taken by the plane indeed follows a great circle that passes through both Beijing and Los Angeles when inspected on a globe. As discussed in class, positive curvature gives rise to elliptic spaces and negative curvature gives rise to hyperbolic spaces. In this project, we will explore what the distance minimizing curves and triangles would look like on spherical and hyperbolic geometries, in contrast to Euclidean geometry. We will derive the formula for areas of triangles in spherical and hyperbolic geometries, and demonstrate how the Gauss-Bonnet formula unifies the triangle area formula in these three situations.

2 Geodesic Triangles on Spheres

Let us first consider spherical surfaces. In the textbook, a geodesic on a curve is defined to be any differentiable curve $\alpha : [a, b] \rightarrow S$ when $\alpha''(t) \perp T_{\alpha(t)}S$ for every $t \in [a, b]$. From example 7.27 from the textbook, we then know that the geodesics of spheres are the great circles. Different from Euclidean geometry where given a straight line and a point (that does not lie on the line) a parallel line could be drawn from it, and where two non-parallel lines must intersect at a single point, on a sphere every great circle intersect with any other great circles exactly twice. However, just as in Euclidean geometry where triangles are defined by three sides, all of which are segments of a straight line, a triangle on a sphere is defined by three great circles as in the figure below.

1. Exercise 16. (Spherical triangles)

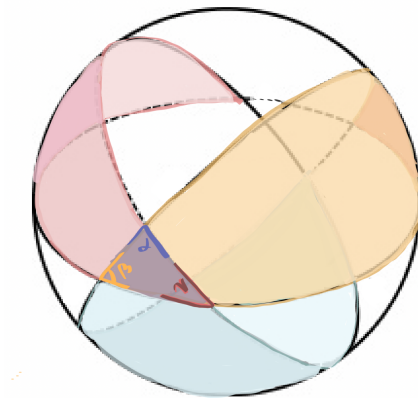


Figure 1: A spherical triangle on a sphere, taken from [1], the cornell DIY document, annotated using procreate

- (a) What is a formula for a triangle with angles α, β, γ on a sphere of radius R ?

Proof. Let W_α be the area of wedge with angle α , and similarly let W_β and W_γ be defined the same way. Also let A be the area of the triangle. As one can see from the figure above, there are two sets of three wedges with the angles α, β , and γ , and each set of wedges intersect at a triangle (with angles α, β , and γ). The two triangles, which have the same area, are each counted three times, resulting an over-counting of A 4 times. Since the total area of a sphere of radius R is $4\pi R^2$, the area of a wedge of angle α is $4\pi R^2(\alpha/2\pi) = 2\alpha R^2$. So the area of the sphere could be described as

$$\begin{aligned} A_{\text{sphere}} &= 2(W_\alpha + W_\beta + W_\gamma) - 4A \\ 4\pi R^2 &= 2(2\alpha R^2 + 2\beta R^2 + 2\gamma R^2) - 4A \\ A &= R^2(\alpha + \beta + \gamma - \pi) \end{aligned} \tag{1}$$

This is precisely the area formula for a spherical triangle with angles α, β , and γ . $\square$

- (b) What angle sums are possible?

Proof. As shown above (1), the area of a spherical triangle is

$$A = R^2(\alpha + \beta + \gamma - \pi).$$

Now since the possible areas of a triangle with area A is $0 < A < 2\pi R^2$ (the upper bound is the area of a half sphere), we get that

$$0 < R^2(\alpha + \beta + \gamma - \pi) < 2\pi R^2,$$

so

$$0 < \alpha + \beta + \gamma - \pi < 2\pi.$$

Adding π to every expression, we get

$$\pi < \alpha + \beta + \gamma < 3\pi.$$

□

This shows that on a sphere the sum of the three angles are strictly greater than 180° , or more precisely, strictly between 180° and 540° . This is different from the restriction on angles in Euclidean space, where the sum of the three angles of a triangle must be 180° . This example also verifies the Gauss-Bonnet formula for geodesic triangles given in class, where sum of the interior angles are

$$\alpha + \beta + \gamma = \pi + \int_T \kappa$$

where T is the triangle on the sphere, and κ is the Gauss curvature on it. Since $\kappa = \frac{1}{R^2}$ is constantly positive on a sphere, we get that

$$\alpha + \beta + \gamma = \pi + \frac{A}{R^2} > \pi.$$

3 Geodesic Triangles on Hyperbolic Disk

Now let us consider hyperbolic geometry. Remember that a hyperbolic plane is a plane where the curvature is constantly negative at every point, where every point is a saddle point. Just like in Euclidean geometry, two distinct points in hyperbolic geometry uniquely defines a line. The hyperbolic plane is often modeled by the Poincaré disk model. The Poincaré's disk model is a unit disk

$$D = \{(x, y) | x^2 + y^2 < 1\}.$$

The boundary of the disk represents points that are infinitely far away from the center. This disk model is equipped with its Poincaré metric tensor:

$$ds^2 = 4 \frac{dx^2 + dy^2}{(1 - x^2 - y^2)^2}.$$

With this metric the Gauss curvature is constantly -1 . The lines (geodesics) on this model are the circular arcs orthogonal to the unit circle and the diameters of the unit circle. The hyperbolic plane has an interesting isometry group. An isometry of the hyperbolic plane is a mapping of the hyperbolic plane to itself that preserves the underlying geometry, including angles and distances. The set of isometries of the hyperbolic plane forms a group. Using complex numbers, we have the following theorem:

Any orientation preserving isometry of the hyperbolic plane T has the form

$$T(z) = e^{i\theta} \frac{z - z_0}{1 - \bar{z}_0 z},$$

where θ is some angle and z_0 is the point inside $\mathbb{D}$ that gets sent to 0.

As in Euclidean geometry, there are three basic isometries of the hyperbolic plane: reflection, rotation, and translation. **Reflection** of the hyperbolic plane are inversions across geodesics. A **rotation** is the composition of two reflections about geodesics meeting at a point P inside H , which is the center of rotation. This isometry resembles an Euclidean rotation as points move around the disk along hyperbolic circles centered at the point P . A **translation** fixes (as a set) the geodesic that connects its two fixed points on the boundary of H . The isometry translates all points in this geodesic by a common distance.

Therefore, we can, without the loss of generality, consider triangles on the hyperbolic plane (on a Poincare disk model) where one of the vertices is at the center of the disk.

As we have verified the Gauss-Bonnet formula for spheres in the previous section, we would like to verify that

$$\sum_{i=1}^3 \theta_i = \pi + \int_T \kappa < \pi,$$

where θ_i are the interior angles of the triangle, and since κ is constantly negative in this case.

We define a one-angle triangle (OAT) of exterior angle α is the region in $\mathbb{H}^2$ cut out by two geodesic rays forming an angle of $\pi - \alpha$ at a point, together with a third infinite geodesic connecting their endpoints, as shown in the figure below:

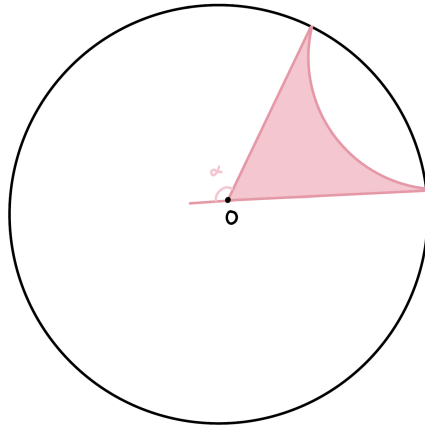


Figure 2: A one-angle triangle with exterior angle α

This could be thought of as taking the limit of a sequence of such triangles:

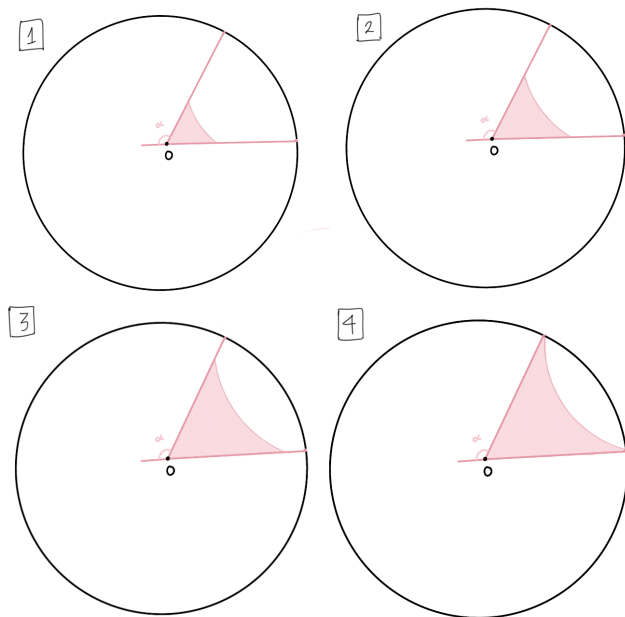


Figure 3: Limit leading to OAT.

Let us denote the area of an OAT with exterior angle α as $A(\alpha)$.

2. Exercise 18. Show that A is additive, i.e. that $A(\alpha) + A(\beta) = A(\alpha + \beta)$. Since A is continuous (in fact, all you really need is that A is monotone – a smaller external angle gives a smaller area), and additive, it must be linear. So there is some constant λ so that $A(\alpha) = \lambda\alpha$. When $\alpha = \pi$, we get a “triangle” with all angles equal to 0. This is called an ideal triangle, and has all vertices on the boundary of the disc.

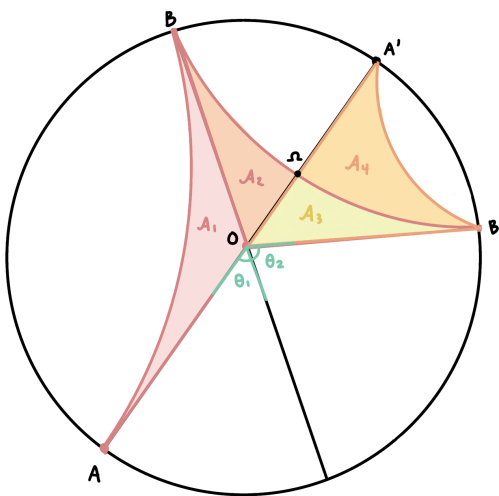


Figure 4: A is additive.

Proof. As shown on the figure above, $A(\theta_1 + \theta_2) = \mathcal{A}_3 + \mathcal{A}_4$. Also, $A(\theta_1) = \mathcal{A}_1$ and $A(\theta_2) = \mathcal{A}_2 + \mathcal{A}_3$. Now notice that if we rotate by π around Ω , ΩAB gets sent to $\Omega A'B'$, so $\mathcal{A}_1 + \mathcal{A}_2 = \mathcal{A}_4$. Then, $A(\theta_1 + \theta_2) = \mathcal{A}_3 + \mathcal{A}_4 = \mathcal{A}_1 + (\mathcal{A}_2 + \mathcal{A}_3) = A(\theta_1) + A(\theta_2)$. Hence A is additive. $\square$

Since A is continuous and additive, it is linear. That is, there exists some λ such that $A(\alpha) = \lambda\alpha$ for every α . When $\alpha = \pi$, we get a triangle whose vertices are on the boundary of the disc, where all interior angles of the triangles are 0. This is called the ideal triangle, and its area in terms of λ is $\lambda\pi$.

3. Exercise 19. Derive a formula for a hyperbolic triangle with angles a, b, c in terms of λ . Note that in particular, you just proved that the angles of a triangle determine the area.

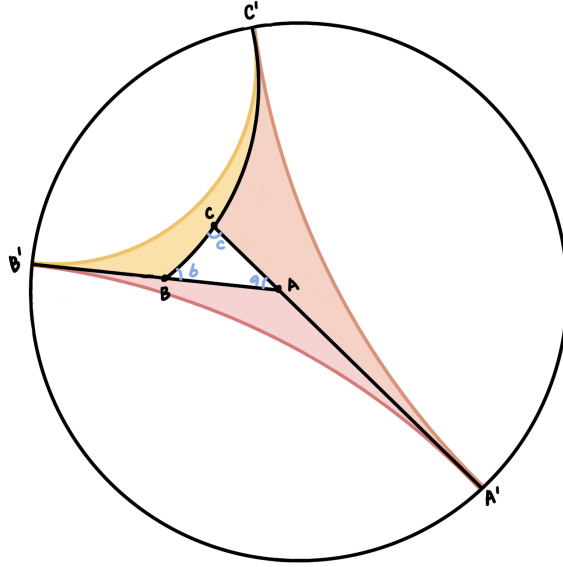


Figure 5: Hyperbolic triangle with angles a, b , and c .

Proof. As discussed before, we can translate point A to the center of the disk while preserving angles and distances. Therefore, without the loss of generality, assume that A is in the center of the disk. We can then extend one end of each of the line segments of the three sides of the triangle to obtain three rays that intersect at the boundary points A' , B' , and C' . This gives rise to the three OATs as shown in the figure above, with exterior angles a, b , and c respectively. Notice that $A'B'C'$ is an ideal triangle, whose area is $\lambda\pi$. We also have, as we derived from exercise 18, that $A(a) = \lambda a$, $A(b) = \lambda b$, and $A(c) = \lambda c$. From the figure above, we can then see that

$$(\text{Area of } ABC) = (\text{Area of } A'B'C') - A(a) - A(b) - A(c),$$

so

$$(\text{Area of } ABC) = \lambda(\pi - a - b - c).$$

□

Note that this also verifies that $a + b + c < \pi$, which we derived from the Gauss-Bonnet formula.

4 Conclusion and Significance

We have thus verified the special case of the Gauss-Bonnet formula of geodesic triangles on spheres and hyperbolic planes. The more general form of the Gauss-Bonnet theorem is as follows:

Theorem Suppose M is a smooth compact two-dimensional Riemannian manifold with boundary ∂M . Let K be the Gaussian curvature of M , and let k_g be the geodesic curvature of ∂M . Then

$$\int_M K dA + \int_{\partial M} k_g ds = 2\pi\chi(M),$$

where dA is the element of area of the surface, and ds is the line element along the boundary of M . Here, $\chi(M)$ is the Euler characteristic of M .

For a polyhedron, the Euler characteristic equals to $F - E + V$, where F is the number of faces, E is the number of edges, and V is the number of vertices. For a surface the Euler characteristic can be defined with triangulations, and it can be proved that $F - E + V$ remains the same for all triangulations. Therefore it is a topological invariant.

In the case of geodesic triangles, we have that $k_g = 0$ along a geodesic, or any side of the triangle. Therefore, we get that

$$\int_T K = 2\pi - \sum_{i=1}^3 \theta_i.$$

In case of a compact two-dimensional manifold without a boundary, the $\int_{\partial M} k_g ds$ could be omitted, and the theorem implies that the total Gaussian curvature of such a closed surface is equal to 2π times the Euler characteristic of the surface. For an orientable compact surface, the Euler characteristic equals $2 - 2g$, where g is the genus of the surface, which counts the number of “holes” of the surface.

This implies that given a compact two-dimensional manifold M without a boundary, since the Euler characteristic remains the same under deformation, the total Gauss curvature enclosed by the surface remains the same. As an application, we then get that since a torus has 0 Euler characteristic, the total Gauss enclosed by it must be zero, even though most points do not have zero Gauss curvature.

Works Cited

All three of the exercises are taken from the DIY hyperbolic geometry document attached to the instruction document.

1. <https://e.math.cornell.edu/people/mann/papers/DIYhyp.pdf>
2. https://en.wikipedia.org/wiki/Gauss%E2%80%93Bonnet_theorem#Interpretation_and_significance
3. https://en.wikipedia.org/wiki/Poincar%C3%A9_disk_model#Metric_and_curvature
4. <https://brice.loustau.eu/ressources/book.pdf>
5. (Graduate Studies in Mathematics, Vol. 69) Sebastian Montiel and Antonio Ros - Curves and Surfaces-American Mathematical Society (2009).pdf
6. [https://math.libretexts.org/Bookshelves/Geometry/Geometry_with_an_Introduction_to_Cosmic_Topology_\(Hitchman\)/05%3A_Hyperbolic_Geometry/5.01%3A_The_Poincare_Disk_Model](https://math.libretexts.org/Bookshelves/Geometry/Geometry_with_an_Introduction_to_Cosmic_Topology_(Hitchman)/05%3A_Hyperbolic_Geometry/5.01%3A_The_Poincare_Disk_Model)